MILITARY SPECIFICATION

CABLE, POWER, ELECTRICAL; AIRPORT LIGHTING, CROSS-LINKED, POLYETHYLENE XLP

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 8 AWG single conductor electric power cable of two voltage ranges intended primarily for use in airport lighting series circuits.

1.2 Classification. Cables shall be of the following classes, as specified (see 6.2):

Class 1 - 3000 V rms

Class 2 - 5000 V rms.

1.2.1 Military part number. Part numbers shall be of the following form, as specified (see 6.2):

Class 1 - M38359 -3K

Class 2 - M38359 -5K

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-343 - Wire, Electrical, Copper (Uninsulated).

MILITARY

MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of U.S. Military Property.

MIL-STD-143 - Specifications and Standards, Order of Precedence for the Selection of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Electronic Support Division (AFLC), 2750 ABW/ES, Gentile AF Station, Dayton, OH 45444 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-810 - Environmental Test Methods.

MIL-STD-831 - Test Reports, Preparation of.

MIL-STD-961 - Military Specification and Associated Documents Preparation of.

MIL-STD-962 - Military Standards and Handbooks, Preparation of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Insulated Cable Engineers Association (ICEA)
(National Electrical Manufacturers Association (NEMA))

ICEA NO. S-19-81 (NEMA WC 3-80) - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

ICEA NO. S-66-524 (NEMA WC 7-82) - Cross-Linked Thermosetting-Polyethylene-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the National Electrical Manufacturers Association 2101 L Street, NW, Washington, DC 20037.)

American National Standards Institute (ANSI)

ICEA NO. S-70-547 (ANSI C 8.35-1975) - Specifications for Weather-Resistant Polyolefin-Covered Wire and Cable.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Cables furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the cable to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Design and construction. The cable shall be designed and constructed for use in airport lighting systems. It shall be constructed for continous outdoor service under all weather conditions while on the surface of the ground, buried directly in the ground, installed in conduit, or submerged in fresh or salt water. The voltage rating, wire size, number of conductors, minimum number of strands, and nominal

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thickness of the combined insulation and shield for the cables specified herein shall be as specified in table I (see 4.6.1).

3.4.1 Conductor. The conductor shall conform to QQ-W-343, type C, class B, no. 8 AWG. It shall be protected by a continuous coating of tin, lead, or lead alloy that will not be affected by the insulation (see 4.6.1).

3.4.1.1 Splices. Splices in the conductor shall be such that each splice will not increase the diameter of the conductor. No splice shall be made in the conductor as a whole. Not more than one strand shall be spliced in any one foot of conductor (see 4.6.1).

TABLE I. Combined cable characteristics.

Characteristic	Class 1	Class 2
Voltage rating	3,000 V rms	5,000 V rms
High potential withstanding voltage	7,500 V rms (5 minutes)	11,000 V rms (5 minutes)
Insulation thickness (average)	0.075 inch (1.91 mm)	0.110 inch (2.79 mm)
Diameter:		
Minimum	0.31 inch (7.87 mm)	0.38 inch (9.65 mm)
Nominal	0.32 inch (8.13 mm)	0.39 inch (9.91 mm)
Maximum	0.34 inch (8.64 mm)	0.41 inch (10.41 mm)

3.4.1.2 Resistance. The resistance of the conductor of 1,000 feet of completed cable at +20° C shall be not greater than 0.65 ohm (see 4.6.8).

3.4.1.3 Conductor shield. A semiconducting layer or tape shall be applied directly over the stranded conductor, over which the cable insulation shall be applied. The conductor shield shall have a minimum thickness of 0.005 inch (0.13 mm) (see 4.6.1).

3.4.1.4 Marker threads. Nonhygroscopic-type marker threads shall be provided in the conductor strands or directly along the conductor for identification purposes. The color and number of threads used shall be identical with those used by the manufacturer on his other products (see 4.6.1).

3.4.2 Insulation. The insulation shall consist of a black cross-linked thermoplastic polyethylene compound which meets the requirements for the insulation included in ICEA S-66-524 and all the requirements specified herein. The insulation shall be applied tightly and concentrically about the conductor over the conductor shield and shall be free from pores, splinters, and other such defects (see 4.6.1).

3.4.2.1 Insulation thickness. The average thickness of the insulation shall be not less than that specified in table I. The minimum thickness at any point shall be not less than 90 percent o the thickness specified in table I. If the insulation is applied in more than one layer, the adjacent layers shall be molded into a homogeneous mass and the mass considered as a whole for all measurements and tests (see 4.6.3.2).

3.4.2.2 Insulation resistance. The measured insulation resistance (see 4.6.3) at 15.6°C, shall be not less than that calculated from the following formula when tested in accordance with ICEA S-19-81 (see 4.6.3):

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$$R = K \log_{10} \frac{D}{d}$$

where R = Megohms 1,000 feet

D = Outside diameter of insulation in inches

d = Diameter of uninsulated conductor in inches

K = Constant 25,000 for insulation specified herein

NOTE: If the temperature differs from 15.6°C during the resistance measurement, correction shall be made by multiplying the measured value by the coefficient corresponding to the temperatures during measurement. The coefficient for the insulation shall be that specified by the manufacturer. The temperature coefficient shall be determined in accordance with ICEA S-19-81.

3.4.2.3 Joints. All joints made in the insulation shall be such that all parts affected in the process will be electrically and physically equal to the remainder of the insulation and shall conform to the thickness requirements specified herein (see 4.6.1).

3.4.2.4 Physical properties. The physical properties of the insulation shall be as follows (see 4.6.9):

a. Original physical requirements:

(1) Tensile strength, initial, 1,800 lbf/in^2 minimum.

(2) Elongation at rupture, 350 percent minimum.

(3) Modules at 200 percent elongation, 1,000 lbf/in^2 minimum.

b. Aging requirements:

(1) After air oven test at 150°C for 168 hours:

(a) Tensile strength in percent of unaged value, 75 percent minimum.

(b) Elongation at rupture in percent of unaged value, 65 percent minimum.

(2) After air pressure heat test for 42 hours 80 lbf/in^2 at 127°C.

(a) Tensile strength in percent of unaged value, 75 percent minimum.

(b) Elongation at rupture in percent of unaged value, 75 percent minimum.

3.4.2.5 Ozone resistance. The cable insulation shall be resistant to ozone attack (see 4.6.6).

3.4.2.6 Moisture absorption. The moisture absorption properties of the cable insulation shall be as follows (see 4.6.5):

Increase in capacitance:

1 to 14 days: 3 percent maximum

7 to 14 days: 1.5 percent maximum.

3.4.2.7 Cold bend. The cable insulation shall not crack as a result of bending at a radius of 2 1/2 times the cable diameter when in an ambient temperature of -40°C (see 4.6.7).

3.5 Performance.

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3.5.1 High temperature. The cable shall be resistant to high temperatures encountered while in storage and under service conditions. When in an ambient temperature of 71°C, a full reel of cable shall not suffer permanent set to any portion of the cable or show evidence of insulation deterioration (see 4.6.10).

3.6 Surface marking on cable. The manufacturer's name and contract number shall be legibly marked on the surface of all cable. The markings shall be of paint or ink applied at intervals not exceeding 10 feet. The markings shall be durable and not readily smeared during handling of the cable. (This paragraph applies only to cable furnished directly to the government under a government contract with the cable vendor as the prime contractor.)

3.7 Workmanship. Workmanship shall be of the quality necessary to produce cable free from all defects which affect proper functioning in service. The exterior surface of the cable shall be smooth, uniform, and free from splinters, ridges, grooves, indentions, and protuberances visible to the naked eye. (This paragraph applies only to cable furnished directly to the government under a government contract with the cable vendor as the prime contractor.)

3.8 Cable lengths. Except as specified in 3.8.1, cable shall be of one continuous 3,000 foot length, +10, -3 percent.

3.8.1 Allowable short lengths. Unless otherwise specified by the acquiring activity (see 6.2), short lengths may be supplied in accordance with the following schedule on orders of 100,000 feet and over:

a. 85 percent or more of the order: 3,000 foot lengths.

b. 10 percent or less of the order: 1,500 foot minimum lengths.

c. 5 percent or less of the order: 1,000 foot minimum lengths.

For orders of less than 100,000 feet, all lengths shall be 3,000 feet +10, -3 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted at room temperature (25°C ±10°C).

4.4 Qualification inspection (see 6.3).

4.4.1 Qualification sample. The qualification sample shall consist of a reel of cable or a 50-foot or longer coil. The sample shall be appropriately identified with the manufacturer's part number and such other information as required by the acquiring activity.

4.4.2 Test report. Upon completion of the qualification inspection, a test report shall be prepared in accordance with MIL-STD-831.

4.4.3 Qualification inspection. The qualification inspection shall consist of all the tests specified under 4.6.

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4.4.4 Retention of qualification. Periodic qualification reevaluations shall be made at three-year intervals after the date of the letter of notification of the product's acceptability for qualification. Materials from current production shall be evaluated against the requirements of 4.5.1 and 4.5.2.

4.4.4.1 Requalification procedure. It shall be the responsibility of the qualified supplier to furnish to the Government, at three-year intervals, the data necessary to establish the continued conformity of the product to all qualification requirements. These data should preferably be complete test results of a sample representative of current production, tested against all the requirements of the specification. At the discretion of the qualifying activity, test records from current production may be accepted for the reevaluation to the extent they are available and samples from current production need be subjected to only the tests for which no production test records are available. The qualifying activity shall be notified of the test results. If a failure occurs, no wire represented by the sample nor any other wire manufactured with the same materials and processes, which has not already been submitted for quality conformance inspection, shall be offered for acceptance until the cause for failure has been determined and concurred in by the qualifying activity as not affecting the ability of the wire to pass qualification inspection requirements. In the event the date for reevaluation has passed and no current production materials or data are available for reevaluation, the supplier shall still be eligible for contract award, but final acceptance of material from such a supplier is contingent upon his material meeting all the qualifying requirements of the specification.

4.5 Quality conformance inspection. The acceptance tests shall consist of the following:

a. Individual tests (see 4.5.1).

b. Sampling plan and tests (see 4.5.2).

4.5.1 Individual tests. Each reel of finished cable shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Alternating-current voltage test (see 4.6.2).

c. Insulation resistance test (see 4.6.3).

d. Continuity test (see 4.6.4).

4.5.2 Sampling plan and tests. Unless otherwise specified (see 6.2), the following tests shall be conducted at least once in every 30-day period, starting at the beginning of production and continuing until completion of the order:

a. Moisture absorption test (see 4.6.5).

b. Ozone resistance test (see 4.6.6).

c. Cold bend test (see 4.6.7).

d. Insulation electric properties test (see 4.6.3.1).

e. Insulation thickness test (see 4.6.3.2).

f. Wire resistance test (see 4.6.8).

g. Physical properties test (see 4.6.9).

h. High temperature test (see 4.6.10).

4.5.2.1 Rejection and retest. When cable selected from a production run fails to meet the specification, no cable still on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately

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corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.1.1 Individual tests may continue. For production reasons, the individual and sampling tests may be continued pending the investigation of a sampling test failure. Final acceptance of cable on hand or produced later shall not be made until it is determined that all cable meets all the requirements of the specification.

4.5.3 Defects in cable already accepted. The investigation of a test failure could indicate that defects may exist in cable already accepted. If so, the contractor shall fully advise the acquiring activity of all defects likely to be found and the method of correcting them.

4.5.4 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-C-12000.

4.6 Test methods.

4.6.1 Inspection of product. All cable shall be inspected to ascertain compliance with this specification with respect to material, workmanship, stranding, and marking.

4.6.2 Alternating-current voltage test. Each reel of cable shall withstand the high potential specified in table II without failure. The test shall be conducted on final shipping reels of cable. For cable supplied on reels containing short lengths of cable, the test may be conducted on reels used for such tests prior to winding on the shipping reel. The high-voltage tests shall be conducted with an alternating-current potential from a transformer and generator of ample capacity, but in no case less than 5 kva. The frequency of the test voltage shall be 60 hertz with a wave shape approximating a sine wave. The initially applied voltage shall be not greater than the rated voltage. The rate of increase shall be approximately uniform and not over 100 percent in 10 seconds nor less than 100 percent in 60 seconds. The reel of cable shall be immersed in a grounded water bath at a temperature of approximately +20°C for 6 consecutive hours and, while still immersed, it shall be subjected to the test potential. The potential shall be applied between the conductor and ground. The duration of the tests shall be 5 minutes.

TABLE II. AC test voltage.

Voltage rating	AWG wire size	Alternating-current test voltage
3,000	8	7,500
5,000	8	11,000

4.6.3 Insulation resistance test. Each reel of cable shall be tested to determine compliance with the insulation resistance requirements specified in 3.4.2.2. This test shall be conducted immediately following the test specified in 4.6.2.

4.6.3.1 Insulation electric properties test. The electric properties of the insulation material shall be tested to determine compliance with 3.4.2.2.

4.6.3.2 Insulation thickness test. The average thickness of the insulation (see 3.4.2.1) shall be determined by measuring the diameter over the insulation and subtracting from it the diameter of the conductor. The difference shall then be divided by two, and the result taken as the thickness of the insulation. Five sets of measurements shall be obtained along the length of a 10-foot sample and the average of the five sets of measurements determined. Each set of measurements shall consist of the determination of the maximum and minimum diameters at the cross-section measured. Measurements shall be made with a micrometer caliper having flat surfaces on both the anvil and end of the spindle, and graduated to read directly in mils, or with a dial micrometer having an anvil, the pressure foot 5/64 inch wide and 3/8 inch long, and exerting a force of 10 grams on the pressure foot. If the results of measurement by such methods are in doubt, the optical method shall be used. The optical method utilizes a micrometer microscope accurate to at least

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0.0002 inch (0.01 mm). Measurements of the diameter shall be made at a section halfway between the ends of the sample and at sections 1 inch (25.4 mm) on each side of the midpoint. The average of the maximum and minimum diameters at each section shall be determined, and the lowest of the three averages shall be used as the diameter of the sample in calculating the thickness of the insulation. The calculated insulation thickness shall meet the requirements specified in table I.

4.6.4 Continuity test. Each length of finished cable shall be tested for continuity of the conductor.

4.6.5 Moisture absorption tests. The moisture absorption tests shall be in accordance with part 6 of ICEA S-19-81. The test temperature shall be 75°C.

4.6.5.1 Surface resistivity (dc potential). The samples shall be subjected to the tests specified in ICEA S-19-81.

4.6.5.2 Surface resistivity (ac potential). Using the same test specimens, similar electrodes spaced 3.00 inches (76.2 mm) apart shall be attached over the central test section. A 60-cycle ac potential increasing uniformly to 20,000 volts in 20 seconds shall be applied between the electrodes and maintained for 1 minute without flashover or damage to the cable.

4.6.5.3 U-bend discharge. The samples shall be subjected to the tests specified in ICEA S-19-81 except the test potential shall be 150 volts per mil of nominal thickness.

4.6.6 Ozone resistance test. Samples of finished cable shall show no injury after exposure for 3 hours to an ozone concentration of not less than 0.025 percent and not more than 0.030 percent (by volume) according to the ozone-resistance tests specified in ICEA S-19-81.

4.6.7 Cold bend test. A sample of the finished cable shall be placed in a cold temperature chamber and maintained at a temperature of -40°C for 24 hours. While still in the cold chamber, the sample shall be given five tight turns around a mandrel whose diameter is five times that of the finished cable. There shall be no visible evidence of cracking. The rate of winding shall be approximately five turns per minute.

4.6.8 Wire resistance test. The resistance of insulated conductor on reels shall be determined at, or the determined resistance corrected to, +20°C and shall be expressed as resistance per 1,000 feet as specified in 3.4.1.2. The following formula may be used for converting resistance measured at temperatures other than +20°C to resistances at +20°C.

$$R_{20} = \frac{254.5\ R_t}{234.5 + t}$$

where: R_{20} = Resistance at +20°C.

R_t = Resistance at temperature (t), in ohms.

t = Temperature at which resistance was measured, in degrees centigrade.

4.6.9 Physical properties test. After aging in an air oven at 150°C for 168 hours, the physical properties specified in 3.4.2.4 shall be tested in accordance with ICEA S-19-81 as follows:

a. Tensile strength (initial and accelerated aging).

b. Elongation (initial and accelerated aging).

4.6.10 High temperature test. The high temperature test shall be conducted in accordance with MIL-STD-810, method 501, procedure I. There shall be no evidence of permanent set or insulation deterioration as a result of this test.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-12000 to the levels specified.

5.2 Identification of product. Each reel of cable shall be marked for identification in accordance wth MIL-STD-130. The following information shall also be applied to each reel of cable:

Voltage rating _____

Size of conductor _____

Number of conductors _____

Length (in feet) _____

Serial no. _____(Reel).

5.3 Marking. Reels shall be marked in accordance with MIL-STD-129. All reels which contain two or more lengths of cable shall be clearly labeled to indicate the number of pieces, the length of each piece, and the location of each length on the reel.

5.4 Cable reels. Cable shall be delivered only on nonreturnable reels.

6. NOTES

6.1 Intended use. The cable covered by this specification is intended for airport series lighting circuits in direct burial or use on the ground.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class cable required (see 1.2).

c. When test conditions are to be other than as specified (see 4.3).

d. If sampling plan is to be other than as specified (see 4.5.2).

e. Lengths to be delivered if other than those specified (see 3.8.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Electronic Support Division (AFLC), 2750 ABW/ES, Gentile AF Station, Dayton OH, 45444.

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible as cited in MIL-STD-961 and MIL-STD-962 within the quality limits required by this document and to fulfill compliance with the resource conservation/recovery act of 1976 (Public Law 94-580) dated 21 Oct 76.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
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Agent:
DLA - ES

Preparing activity:
Air Force - 85

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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